U.S. Police Line-of-Duty Deaths

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Abstract:

U.S. law enforcement line-of-duty deaths have been recorded since 1791, but most research on them is descriptive, limited to felonious killings, or based on aggregated counts. This project uses a Kaggle dataset of 26,638 rows of death records from 1791 to 2022, originating from the Officer Down Memorial Page, to ask how well officer age, length of service, rank, timing, and location predict the cause of death and the offender's legal outcome. In Semester 1, we will clean your data, compare three machine learning classifiers against a baseline using macro-F1, and use Poisson and negative binomial regression to analyze officer deaths per state and year using FBI, BJS, and BLS counts. In Semester 2, we will use natural language processing to analyze the summaries and track how themes change over time. The deliverables are a cleaned dataset, a reproducible code repository, a dashboard, and a final report.

Introduction:

Background Information:

Line-of-duty deaths are among the most serious outcomes in public safety. They harm officers' families, their departments, and the communities they serve. In 2025, 53 officers were feloniously killed in the line of duty, and 81.1% of them died from gunshot wounds.[1] Research shows that lethality depends on situational and encounter characteristics[2], on officer age and length of service [3], and on community-level conditions such as law enforcement presence and training [4]. Knowing which officer and organizational characteristics go with which kinds of fatal outcomes can help agencies set training priorities, plan scheduling, and allocate resources.

Research Gap / Challenges:

Most existing work on this problem is descriptive or regression-based. It typically examines felonious killings only, mostly gunfire, and often uses aggregated counts or short time windows. One national study pooled cross-sectional data to estimate the odds that an assault turns lethal [2]. Another revisited the effects of tenure and age on felonious deaths from 1995 to 2015 [3], and a third analyzed contributing factors at the county level [4]. Supervised machine learning is used in policing, but mostly for other targets. In Charlotte-Mecklenburg, a machine learning model flagged more at-risk officers, with fewer false positives, than the department's

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[1] Federal Bureau of Investigation, "FBI Releases Officers Killed and Assaulted in the Line of Duty, 2025 Special Report and Law Enforcement Employee Counts," May 11, 2026. fbi.gov.

[2] Assaults against U.S. law enforcement officers in the line-of-duty: situational context and predictors of lethality," 2016. PubMed PMID: 27885587. https://pubmed.ncbi.nlm.nih.gov/27885587/

[3] K. Tucker-Gail et al., "Felonious line-of-duty officer deaths (1995–2015): The impact of tenure and age revisited," 2022. doi: 10.1177/14613557211032608.

[4] J. M. Doucet and L. A. Eargle, "Law Enforcement Murdered in the Line of Duty: A County-Level Analysis of Contributing Factors," Criminology, Criminal Justice, Law & Society, vol. 24, no. 2, pp. 9–30, 2023.

existing early-intervention system.[5] That model predicts adverse events involving officer conduct, not officer fatalities. To our knowledge, no study uses officer age, length of service, rank, day of week, and department type to jointly classify (a) the cause or weapon involved in the death and (b) the offender's legal outcome. Few studies also combine this with death rates per officer by state and year, or with text analysis of incident narratives.

The data raise several challenges:

- Missing values. Age is missing for 10.4% of records, and the offender field is missing for 46.1%.
- Free-text offender field. It contains values such as "Executed in 1791" and "Acquitted," so it must be grouped into consistent outcome classes.
- Class imbalance. Gunfire dominates the cause categories, so we will use grouped targets, class weights, and macro-F1 evaluation.
- Limited predictors. These variables capture only part of what determines a death, so model performance may be modest.
- Rates need outside data. Officer counts by state come from other sources and are available only from roughly the 1990s to 2022.
- Unusual events. COVID-19 and 9/11 produced large numbers of illness-related and multi-officer deaths that would distort rates if left unhandled.

Objective(s):

Our main goal is to understand how officer-level and organizational factors relate to line-of-duty deaths in the U.S. We will do this through supervised classification, death-rate modeling by geography and force size, and text mining of the incident summaries.

[5] Officer Down Memorial Page, official website and historical memorial records. https://www.odmp.org/ Accessed Oct. 5, 2026.

Semester 1: Data preparation, classification, and death-rate analysis

1. Prepare and audit the data. Clean and standardize the records, and group the free-text offender field into consistent legal-outcome classes. Label each death by cause type (violent, accidental, or illness). Flag multi-officer incidents, such as 9/11, so one event is not treated as many independent cases. Verify the cleaned row count and the gunfire class percentages before modeling.
2. Build and compare classification models. Train decision tree, random forest, and XGBoost models against a majority-class baseline to predict (a) cause of death and (b) offender legal outcome. We will use a split with cross-validation on the training data. The primary metric is macro-F1, and we will report whether each model beats the baseline on the held-out test set.
3. To analyze officer death rates from 1990 to 2022, you will merge death records with FBI, BJS, and BLS officer counts, run Poisson and negative binomial regressions using headcount as an offset and location, year, and agency size as predictors, and then pick the best model using dispersion and AIC to report Incidence Rate Ratios (IRRs) with 95% confidence intervals.
4. Use BJS federal officer counts to compare federal and state death rates per officer using a rate-ratio test at a 0.05 significance level
5. Instead of dropping anomaly years, we will calculate death rates separately by cause type so that the COVID-19 and delayed 9/11 illness spikes only affect the illness-specific baseline.

Dataset:

Type of Data

The dataset (police_deaths_USA_v7.csv) is a tabular CSV file with 26,638 rows and 13 columns. Each row is one U.S. law enforcement officer who died in the line of duty, and the records run from January 3, 1791 to December 4, 2022. The columns are a mix of types:

- Numeric: Age.
- Date: End_of_Watch, which has mixed date formats that we parse during cleaning.
- Categorical: Rank, Day_of_Week, Cause, Department, State, Weapon, and Offender.
- Free text: Tour, Badge, Name, and Summary. Tour is stored as text (e.g., "19 years, 6 months") and is converted to a numeric Tour_Years feature. Summary holds a narrative of the incident for each officer. It is the basis of the Semester 2 text analysis.

For this project, we exclude Name, Badge, and the narrative Summary from the supervised models.

Origin / Source

The dataset was downloaded from Kaggle as police_deaths_USA_v7.csv [Mayuresh Koli, "Police deaths in USA from 1791 to 2022," Kaggle, last updated in 2022. https://www.kaggle.com/datasets/mayureshkoli/police-deaths-in-usa-from-1791-to-2022/data. Accessed September 9, 2026.[6] The records originate from the Officer Down Memorial Page (ODMP), a nonprofit database begun in 1996 that documents U.S. law enforcement officers who died in the line of duty [5].

[6] M. Koli, "Police Deaths in USA from 1791 to 2022," Kaggle, [year]. [Online]. Available: https://www.kaggle.com/datasets/mayureshkoli/police-deaths-in-usa-from-1791-to-2022. Accessed: [Sept. 9, 2026].

Size and Distribution

- Total number of samples (rows for tabular data, images for computer vision, sequences for time-series).
- There are a total of 25,638 rows and 13 columns:
    - Rank: Constable, Sheriff, Marshall, etc.
    - Name: Darius Quimby, Isaac Smith, Robert Berwick
    - Age: 53, 40, 45
    - End_Of_Watch: 1791-01-03, 1792-10-22, 1797-06-29
    - Day_Of_Week: Monday, Saturday, Thursday
    - Cause: Stabbed, Gunfire, Duty-related
    - Department: Albany County Constable's Office, New York
    - State: New York, United States, Texas
    - Tour: 4 years, 4 months; 2 years; 1 year, 4 months
    - Badge: 4, Superintendent
    - Weapon: Edged weapon; knife, Handgun, Handgun; Pistol
    - Offender: Escaped, Executed in 1791, Acquitted
    - Summary: Constable Darius Quimby was stabbed to death w...

Data Quality and Challenges

- There are missing values for Badge, Offender, Weapon, Tour, Age, and one missing value for Cause.
- As for outliers, there are some in the Age category
- There is a lot of inconsistent labeling in the Badge and Rank categories.

- Since the dataset is large and there is a lot of missing information for the older dates, we may have to subset the data to get a better result.

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Dataset Example / Screenshot

Rank	Name	Age	End_Of_W	Day_Of_W	Cause	Department	State	Tour	Badge	Weapon	Offender	Summary
Constable	Darius Quimby		1791-01-0	Monday	Stabbed	Albany County Constable's Office, N	New York			Edged wea	Executed i	Constable Da
Sheriff	Cornelius Hogeboom	53	1791-10-2	Saturday	Gunfire	Columbia County Sheriff's Office, Ne	New York	2 years		Gun; Unkn	Acquitted	Sheriff Corne
Deputy She	Isaac Smith		1792-05-1	Thursday	Gunfire	Westchester County Sheriff's Depar	New York			Handgun	Executed i	Deputy Sherif
Marshal	Robert Forsyth	40	1794-01-1	Saturday	Gunfire	United States Department of Justice	United Stat	4 years, 4 r	4	Handgun; I	Escaped	Marshal Robe
Deputy She	Robert Berwick		1797-06-2	Thursday	Gunfire	New York County Sheriff's Office, Ne	New York			Handgun	Executed i	Deputy Sherif
Sheriff	Robert Maxwell	45	1797-11-1	Sunday	Gunfire	Washington District Sheriff's Office,	South Carolina			Gun; Unkn	Acquitted	Sheriff Robert
Superinten	Hilary Baker	52	1798-09-2	Tuesday	Duty relate	Philadelphia Police Department, Pe	Pennsylvar	1 year, 4 m	SUPERINTEN			Superindente

Methodology:

In this section, we outline the step-by-step approach we will follow for both Capstone 1 (predictive modeling) and Capstone 2 (narrative analysis). We will model deaths on a state-year plane in which every death is split into a cause-type group (such as violent, accidental, illness, etc.) and the number of officers in each state as an input feature. We will create a new cause-type feature for each row so we can learn how different kinds of deaths relate to officer counts, year, and region. Capstone 1 will build and compare models on the structured data while Capstone 2 will include the features extracted from the summaries.

Data Understanding and Exploration

First, we will perform exploratory data analysis (EDA) to understand the dataset. Since the records span from 1700 to the 20th century, we expect differences in quality and wording, and EDA will help us document those differences. We will count records and columns, note their data types, and count unique values. Then we will group the recorded causes into violent (gunfire, stabbing, assault), accidental (automobile/vehicular incidents), and illness (heart attack, COVID-19, 9/11-related illness), and report the number of deaths for each group.

We will also obtain a summary of officers by state and year from the FBI, LEOKA, and UCR programs, as well as the BJS Census of State and Local Law Enforcement Agencies, and their rank counts by state.

We will use histograms for age, a line chart for cause of death every 50 years for each cause-type group, and a bar chart for the top 10/15 causes of death by state and rank. We will plot deaths against officer counts by state, then compare them by cause type across regions.

One of the main challenges we will face is the number of missing values per column. Based on the Excel file, we can see many missing values in the badge, offender, weapon, tour, and age columns. We can expect noise from multi-officer incidents, which can cause spikes, especially during major historical events such as 9/11 and COVID-19. We may have issues with officer counts for data before the 1900s, and the actual amount may vary by source, which could limit the model panel to recent years.

Data Preprocessing and Preparation

The following steps will be applied to the Capstone 1 dataset and used for the summaries in Capstone 2.

Cleaning:

We will standardize column names, remove duplicate rows (if any), and convert ages to numeric values. For any unusual ages, we will search for the officer's actual age online and manually correct them rather than deleting them.

For cause-type grouping, we will assign each cause to broader groups, and for uncommon/singular causes, we can group them into an ‘Other’ category. We will then count the deaths for every combination of state/federal, year, and cause type. Lastly, for the officer count, we will merge officer counts into a panel, take the log to reduce skew, and compute officers per 100,000 residents using Census population estimates.

Missing Values:

For missing values, we will handle them based on their usefulness to the project. Columns that are mostly empty or don't carry any predictive meaning will most likely be dropped. For the ones that remain, we will fill numeric categories with the median and categorical variables with the most frequent category. We will fit all imputation on the training data and then apply it to the test data to avoid data leakage.

Encoding and Standardizing:

For the categorical variables (cause type, region, and state), we will use one-hot encoding and as for the numeric variables (year, log officer count, and agency size), we will standardize them for the Poisson and negative binomial regression models. The tree models won't require any standardization, but we will still use the same inputs for all of the models.

Class Imbalance and Augmentation:

We will use loss functions designed for count data to predict the number of deaths. We will also consider two separate predictions: first, whether any deaths occur, and second, how many deaths occur. If we classify state-year observations into low and high death rates, we can use class weights or oversampling during model training to help address imbalances between the two groups.

Modeling Plan

We will model a state-year-cause-type panel, with officer count and cause type as features.

- Planned Models:

- Baseline – the average death count for each cause type
- Poisson regression and negative binomial regression – statistical count models with officer count as the ordinary predictor
- Decision Tree – single tree with limited depth
- Random Forest – multiple trees
- XGBoost – gradient boosting using the Poisson for counts

- Justification:
  - Since each row represents one state, year, and cause type, and our target is the number of deaths for each row, many zeros are likely because some years/states have no deaths for a specific incident. Therefore, using Poisson and negative binomial will help handle counts that vary more than usual.
  - Tree models can capture interactions between categorical and numerical variables and nonlinear patterns.
  - Random Forest is stable with smaller datasets, and XGBoost will hopefully perform better and can also support Poisson loss.
- Implementation Tools:
  - We will use Python with:
    - Pandas and NumPy for data handling
    - Scikit-learn for pipelines, trees, and random forest
    - Statsmodels for Poisson and negative binomial regression
    - XGBoost
    - Matplotlib for plots
    - Google Colab

        - Jupyter

Model Training Strategy

- Splitting
    - We will split by time instead of randomly. As of right now, there isn't a specific split for each group, but we will probably have training be the older years, then validate on the middle ranges, and test on the more recent years.
- Number of epochs, batch size (initial suggestions).
    - For XGBoost, we will start with probably up to 500 boosting rounds with early stopping on the validation set and a learning rate of 0.05 to 0.1 with depth between 3 and 6.
    - For Random Forest, we will probably start with 250 to 500 trees.

Evaluation Plan

- Regression Metrics:
    - We will report the mean absolute error (MAE), root mean squared error (RMSE), R-squared, and Poisson deviance on the test years. We will then compare each of the models with the baseline
- Classification Metrics:
    - We can also report the accuracy, precision, recall, F1-score, a confusion matrix, and a ROC curve.
- Error Analysis:
    - We will look at the largest prediction errors by state, year, and cause type, with attention to multi-officer incidents and small states that don't have as many deaths.

Plan for Capstone 2 (Next Semester)

We will use the Capstone 1 error analysis to refine the features and retune the models. Then we will use text analysis to identify circumstances missing from the structured data, such as traffic stops, warrant services, and ambushes. Since almost half of the offenders outcomes are missing, we will use the provided summaries to infer the outcome, then validate the method on records where the outcome is known.

We can also compare the causes and circumstances of federal and state officer deaths using the summaries and a chi-squared test at a 0.05 significance level. We will also implement Random Forest, XGBoost, and negative binomial models, and we will explore additional models to improve the analysis.

Expected Outcomes

Applications

This project is intended as an evidence base for officer-safety research and policy, not as a tool for predicting which individual officer will be harmed. Predictors such as age, length of service, rank, and day of the week describe only part of what determines a death, so we expect modest model performance and will say so plainly. If certain tenure, rank, or time-of-week patterns are linked to particular causes of death, agencies and training academies could use the results to decide where to focus safety training and refresher courses. Such findings would be hypotheses to test, not conclusions to act on alone. Descriptive findings by state and time period could help departments and policymakers see where and when fatal risk has historically been concentrated, and how it has shifted over time. The cleaned dataset, the grouped offender legal-outcome classes, and the baseline results give later researchers a documented starting point for this problem. As noted in the research gap, few studies model the cause of death and the offender's legal outcome together. The data are historical and span more than two centuries, and many records have missing fields. Reporting is aggregate only, and we will state the limits of the data clearly so the results are not read as predictions about specific officers or departments.

Deliverables

We plan on producing a cleaned dataset and data dictionary. The cleaned dataset will have standardized dates and a numeric tenure variable. It will also have documented offender legal-outcome classes, and the data dictionary will define every variable and each cleaning decision. Reproducible code repository. A GitHub repository will contain the full pipeline: preprocessing, baseline, decision tree, and random forest models, the fixed train/test split, and the cross-validation setup. It will also include a README with run instructions so anyone can

reproduce our results. Model comparison and results. We will report macro-F1 and related metrics on the held-out test set for both prediction targets, with feature-importance results and diagnostic plots. Visualization report or dashboard. A Tableau dashboard or report will show the trends by cause, time period, state, and officer characteristics, built from the Semester 1 EDA. Semester 2 text-analysis component. We will apply topic modeling (NMF and LDA) to the narrative Summary field and report how incident themes have changed over time. Final report and presentation. A final written report and presentation will cover the full workflow, results, and limitations.

Team Plan

Our team has four members: Jayden Howard, Jennifer Tran, Ruth Slaughter, and Devashish Patel. Each member leads one stage of the pipeline and also reviews the work of the stage before and after theirs. Every member is responsible for the accuracy of the full project, so each will be able to explain and reproduce any part of the analysis.

- Jayden Howard, data audit and panel construction. Audit missing values and inconsistent fields (Age, Tour, Offender, mixed date formats). Map causes to the violent, accidental, illness, and Other groups, flag multi-officer incidents, and build the state-year-cause-type panel including zero-count cells.
- Jennifer Tran, Exploratory analysis and state context: Produce the EDA on cause distribution, age, weekday, geography, and long-term trends. Collect and merge officer counts and population data from FBI, BJS, BLS, and Census sources, and build the Tableau dashboard.
- Ruth Slaughter, baseline and count models. Build the preprocessing pipeline, baseline, Poisson and negative binomial models, and time-based split.
- Devashish Patel, tree models, evaluation, and model selection. Build and tune the decision tree, random forest, and XGBoost models. Evaluate all models against the baseline, and produce the diagnostic plots and feature importance results

Shared work. All four members will take part in the following:

- Agreeing on the final cause-type mapping, panel definition, and split years, including any changes after instructor feedback
- Reviewing each other's code and results before they are reported
- Writing the final report and presentation.

[1] Federal Bureau of Investigation, “FBI Releases Officers Killed and Assaulted in the Line of Duty, 2025 Special Report and Law Enforcement Employee Counts,” May 11, 2026. fbi.gov.

[2] “Assaults against U.S. law enforcement officers in the line-of-duty: situational context and predictors of lethality,” 2016. PubMed PMID: 27885587. https://pubmed.ncbi.nlm.nih.gov/27885587/

[3] K. Tucker-Gail et al., “Felonious line-of-duty officer deaths (1995–2015): The impact of tenure and age revisited,” 2022. doi: 10.1177/14613557211032608.

[4] J. M. Doucet and L. A. Eargle, “Law Enforcement Murdered in the Line of Duty: A County-Level Analysis of Contributing Factors,” Criminology, Criminal Justice, Law & Society, vol. 24, no. 2, pp. 9–30, 2023.

[5] Officer Down Memorial Page, official website and historical memorial records. https://www.odmp.org/ Accessed Oct. 5, 2026.

[6] M. Koli, “Police Deaths in USA from 1791 to 2022,” Kaggle, [year]. [Online]. Available: https://www.kaggle.com/datasets/mayureshkoli/police-deaths-in-usa-from-1791-to-2022. Accessed: [Sept. 9, 2026].